Inborn Errors of Metabolism 1

Dr Gada Jadkarim

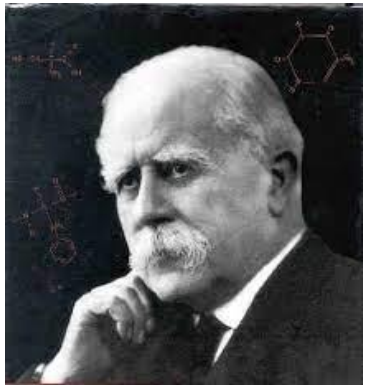
Clinical and Biochemical Geneticist

King Abdul-Aziz University

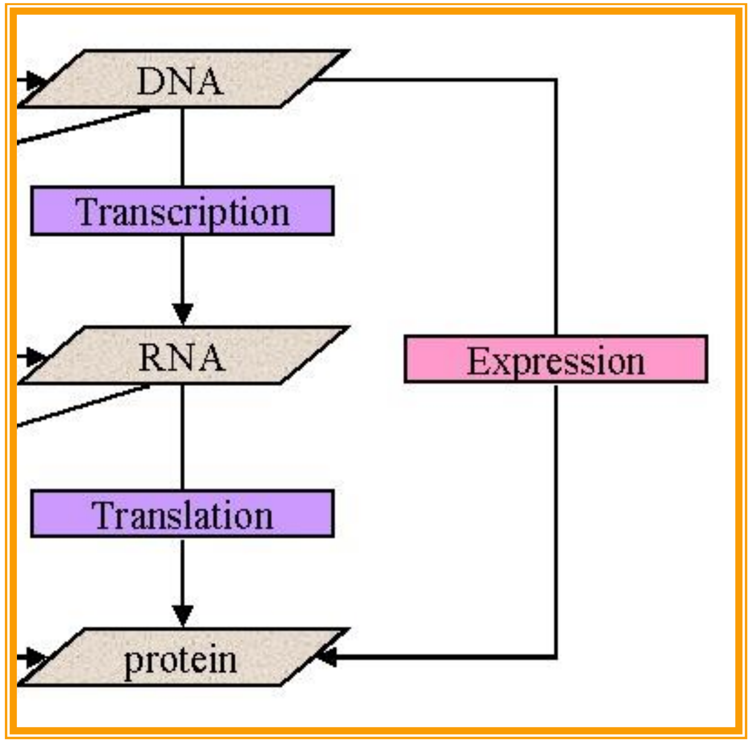
Learning Outcomes:

1. Define inborn errors of metabolism and their clinical implications.
2. Demonstrate an understanding of various metabolic disorders and their underlying mechanisms.
3. Identify the clinical presentations and symptoms associated with metabolic disorders.
4. Discuss the management strategies for metabolic diseases and their importance in patient care.

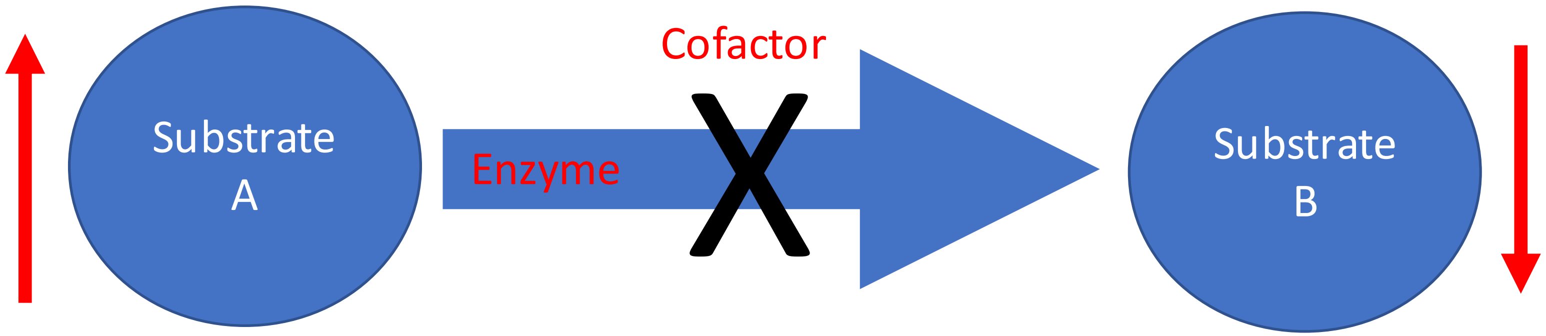
Definitions



- Metabolism
  - The chemical reaction constituting the process of breakdown and synthesis.
    - Enzymes play an indispensable role in facilitating the process.
- Sir Archibald Garrod introduced the term "inborn errors of metabolism"
  - Inherited enzyme defects



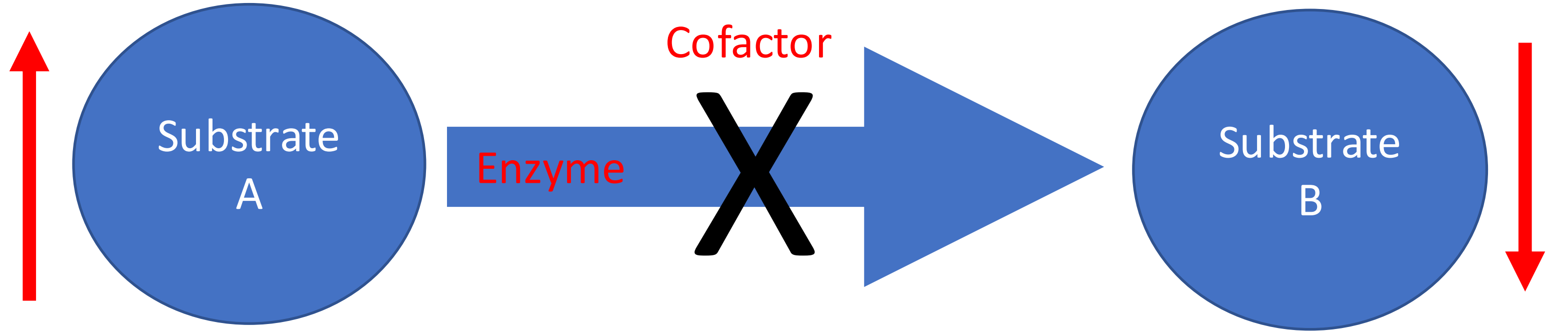
Mechanism of IEM



- Accumulation of a toxic metabolite
- Deficiency of down stream metabolites

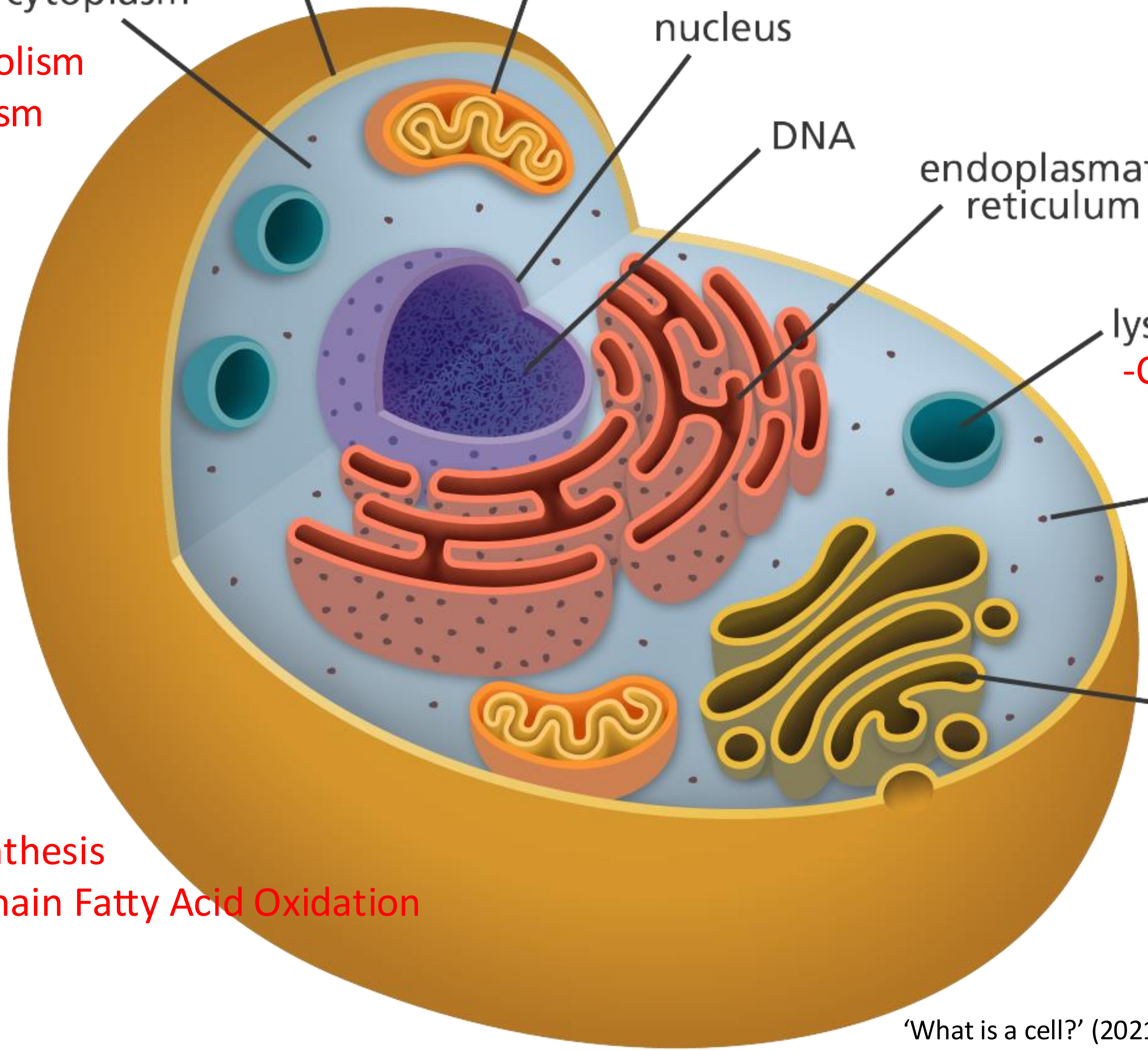
Major types of IEMs

- Disorders of intoxication
- Disorders of Energy metabolism
- Disorders of Complex Molecules



-Carbohydrate metabolism
-Amino acid metabolism
-Distal urea cycle

-Fatty Acid metabolism
-Oxidative phosphorylation
-Proximal urea cycle



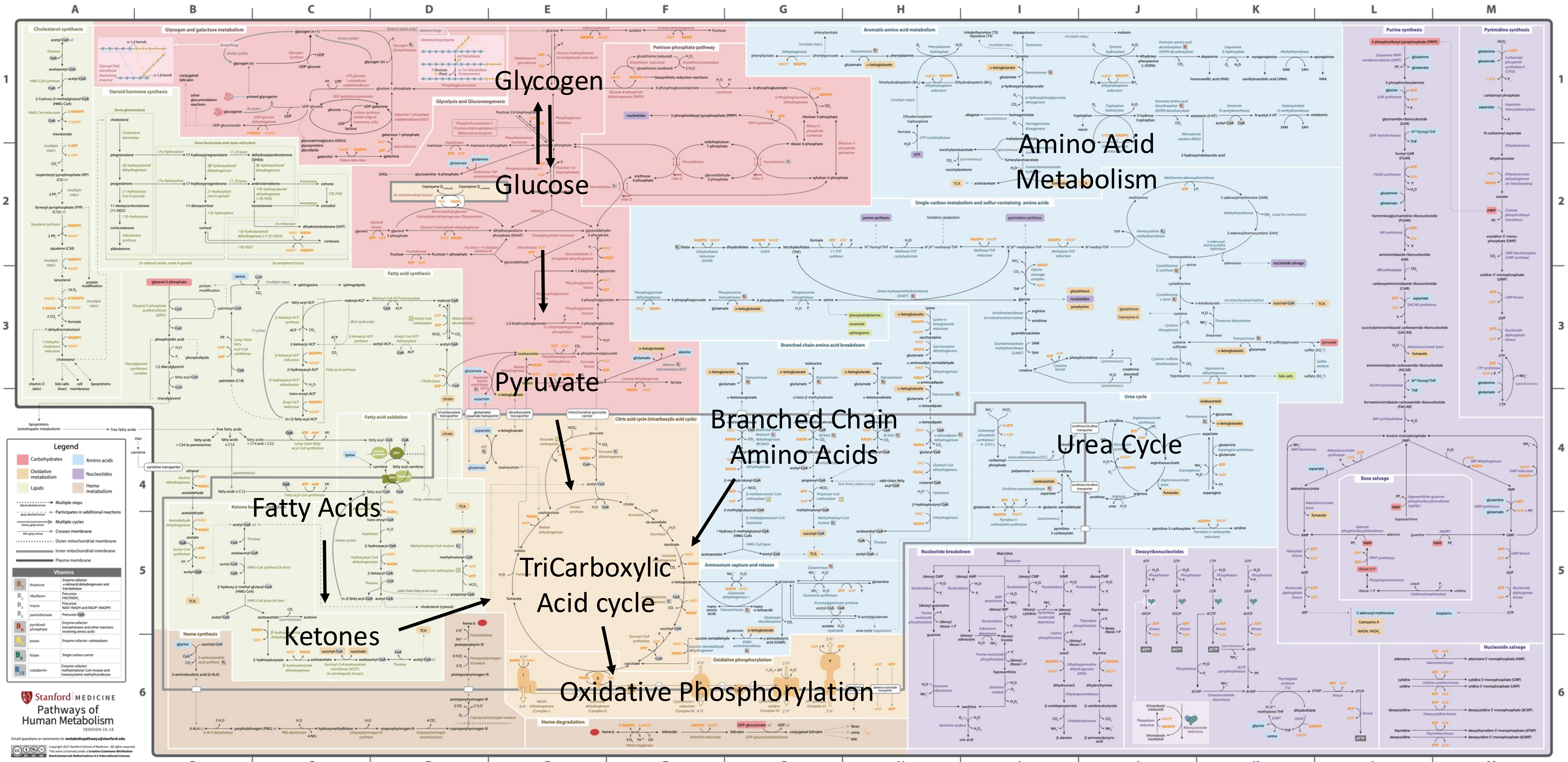
-Glycosylation (endoplasmatic reticulum)

-Complex molecule breakdown (lysosome)

-Glycosylation (Golgi apparatus)

Peroxisomes
-Bile Acid Synthesis
-Very Long Chain Fatty Acid Oxidation

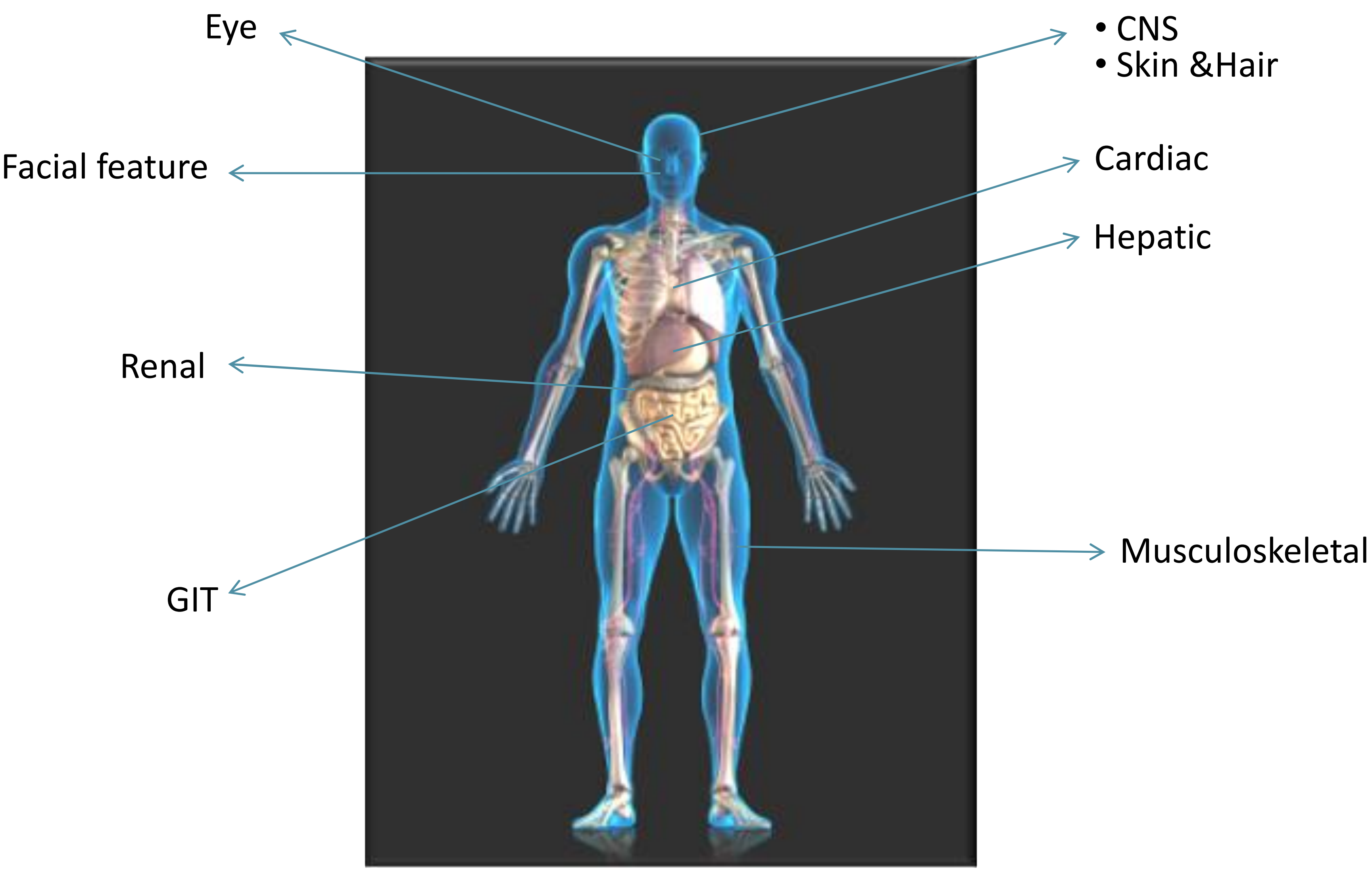
‘What is a cell?’ (2021) Your Genome. Available at: https://www.yourgenome.org/facts/what-is-a-cell (Accessed: 5 January 2021).



”Pathways of Human Metabolism (2018) version 10.18. Available at: https://metabolicpathways.stanford.edu/ (Accessed 5th January 2021)

Presentation of IEM

Fetal Neonatal infancy Childhood Adolescence Adulthood



- Broad clinical manifestations
  - However, we do see several themes to run consistently.

Examples of IEM Disorders

Disorders of intoxication

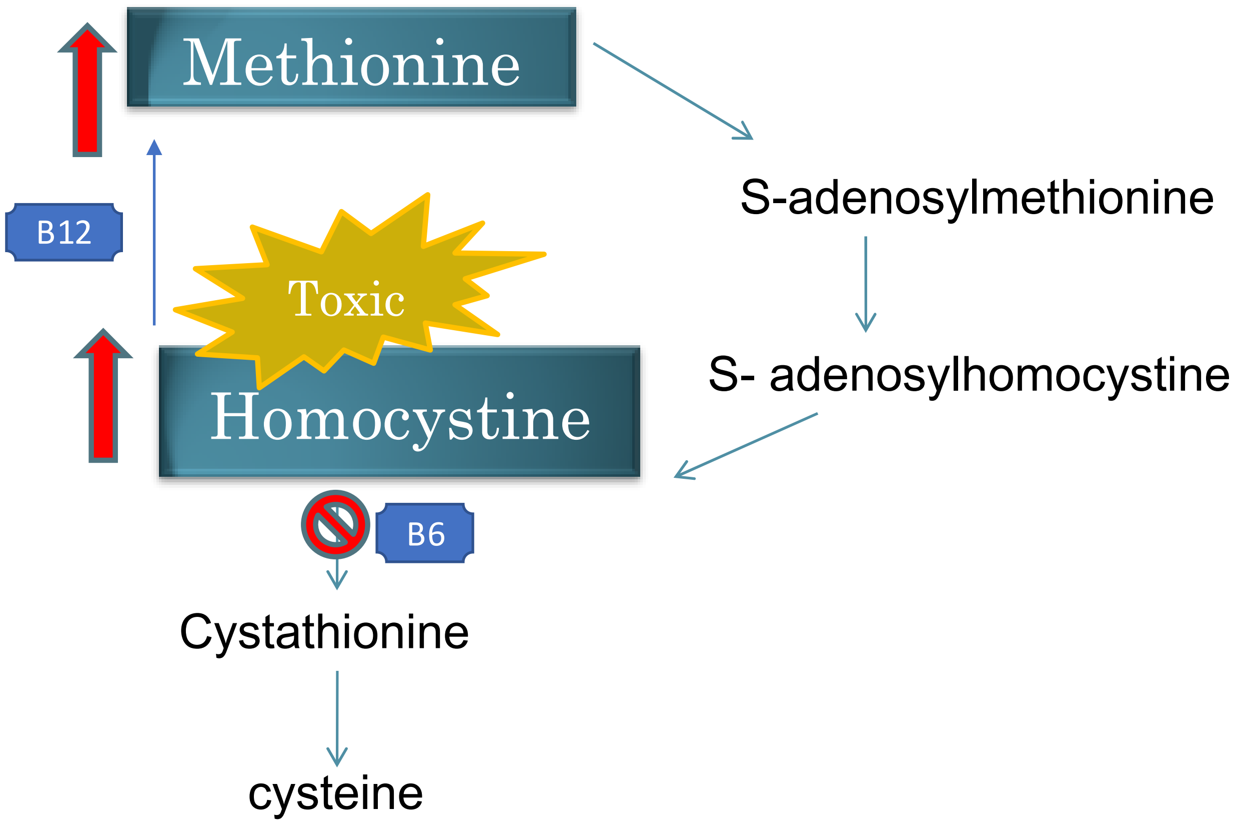
- Aminoacidopathies
  - **Phenylketonuria (PKU)**
  - **Homocystinuria**
  - **Maple Syrup Urine Disease (MSUD)**
- Urea Cycle defects
  - **Ornithine Transcarbamylase Deficiency (OTCD)**
- Organic Acidemias
- Sugar intolerance
  - **Galactosemia**
- Metals and Porphyria

Case

- 11 years old boy presented with:
  - Intellectual Disability
  - Seizure
  - Poor vision
  - Consanguinious parents
- Examination
  - Tall with thin long limbs
  - Lens subluxation
  - Scoliosis
- Testing:
  - Plasma Amino Acids
    - Increased Methionine
  - Plasma Homocysteine
    - Increased

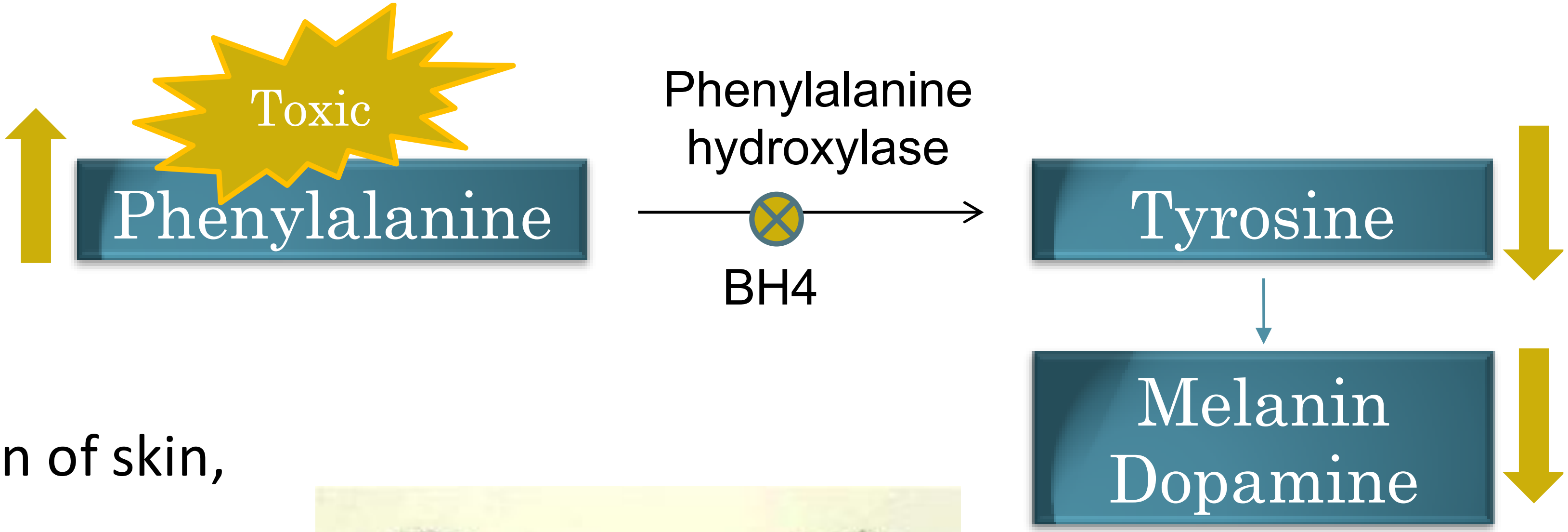
Homocystinuria

- Due to:
  - Cystathionine Beta Synthetetase deficiency
  - Vitamin B12 Deficiency
    - Cobalamin disorder
    - Dietary Deficiency

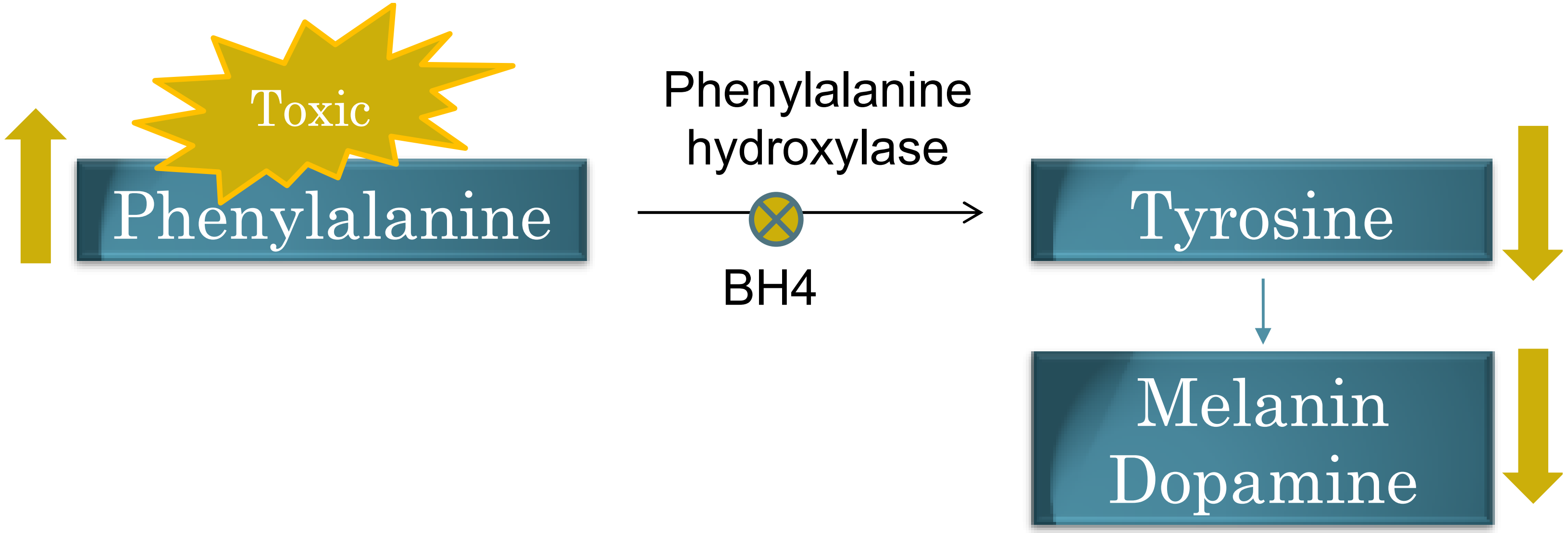


Amino Acid Metabolism – Phenylketonuria (PKU)

- Autosomal Recessive
- Clinical features
  - Intellectual Disability
  - Seizures
  - Decrease pigmentation of skin, hair and iris
  - Abnormal urine odor (mousy)
- Almost all cases are identified through newborn screen



PKU - treatment



- Diet
  - Low phenylalanine, tyrosine rich
- Cofactor

Case

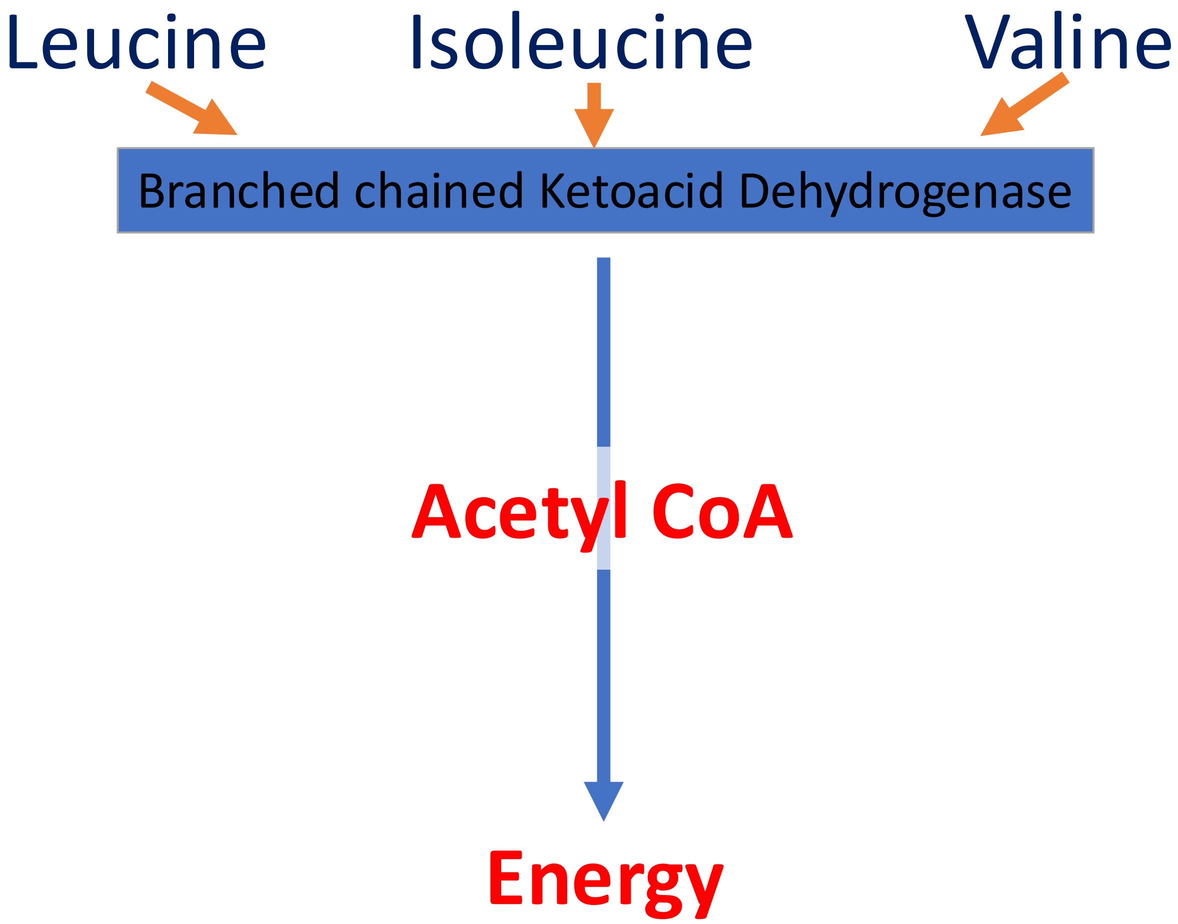
- One week old baby presented with:
  - Vomiting, poor feeding and sucking
  - Decrease in his activity and seizure
  - The urine smell was not normal?

- His Newborn screening showed :
  - Increased Leucine, Isoleucine, and Valine

Maple Syrup Urine Disease

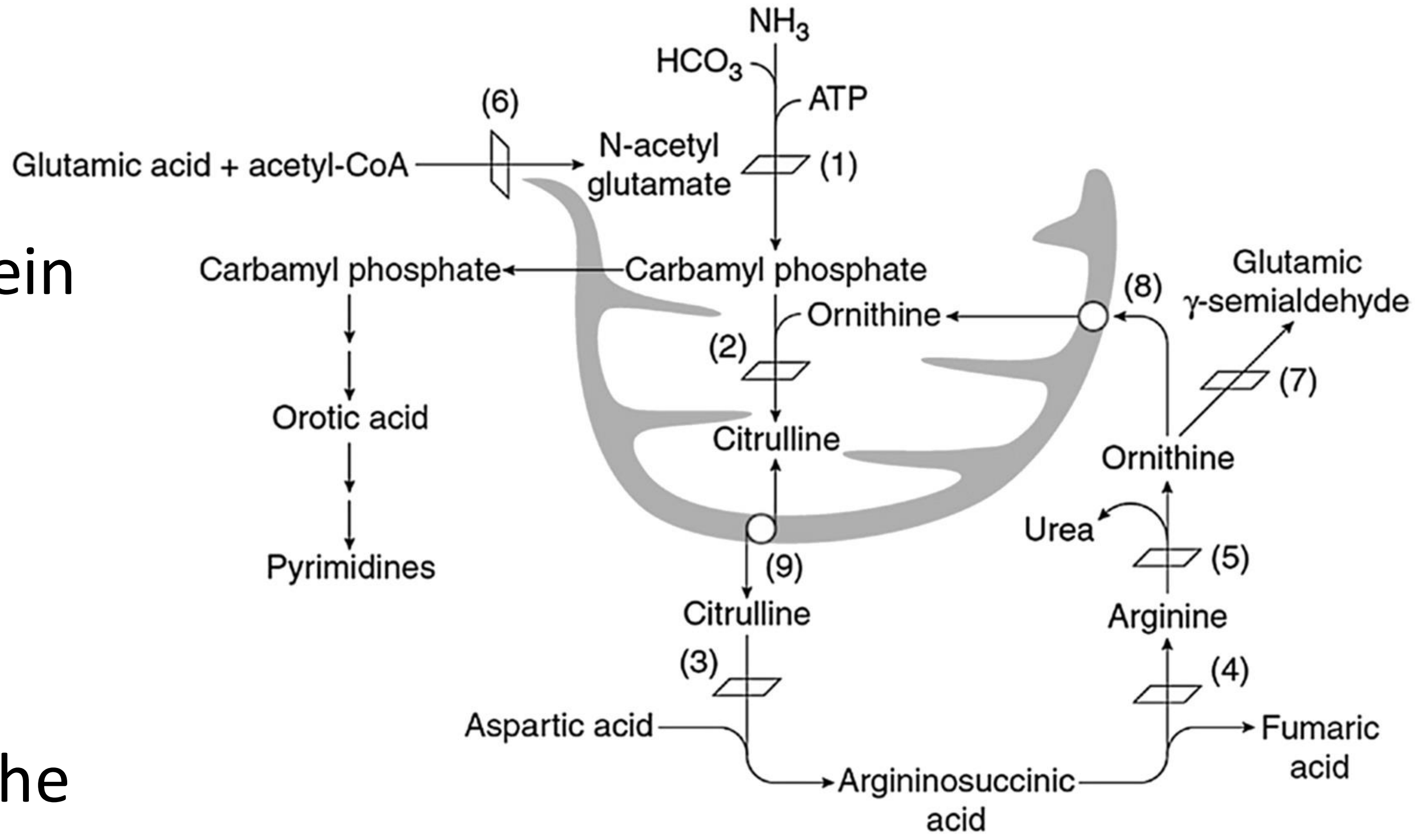
- Plasma Amino Acids
  - Increase alloisoleucine, isoleucine, leucine, valine

- Treatment
  - Restriction of Branched Chain Amino Acids (Leucine, isoleucine, and valine)
  - Thiamine (Vitamin B1)
  - Liver Transplant



Urea Cycle (UC)

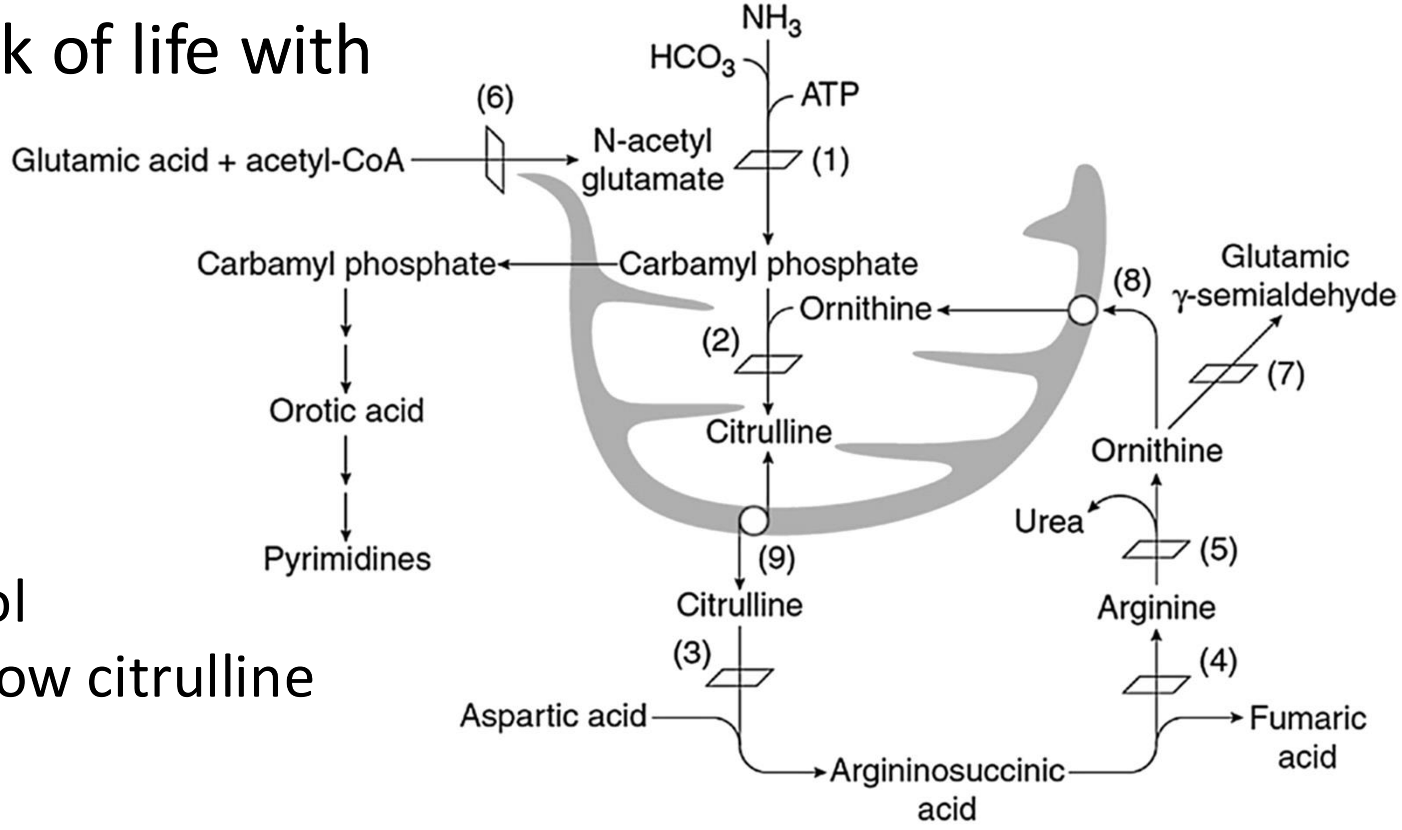
- Nitrogen is formed from protein turnover → Ammonia
- High ammonia result in CNS toxicity.
  - Increased uptake to astrocytes
    - Leading to cerebral edema.
- UC is the major pathway for the removal of nitrogen.
  - Occurs primarily, in the liver.



*Sanchez Russo RL & Wilcox WR.(2021) Amino Acid Metabolism. In Pyeritz RE et al. Emery and Rimoin's Principles and Practice of Medical Genetics and Genomics: Metabolic Disorders. Elsavier. 49-104

Ornithine TransCarbamylase Deficiency (OTCD)

- X-linked disorder
- Presents during the 1st week of life with Hyperammonemia
  - Lethargy
  - Poor feeding
  - Coma
  - Death
- Investigation
  - Hyperammonemia >400 umol
  - Elevated plasma glutamine, low citrulline
  - Elevated Urine Orotic acid



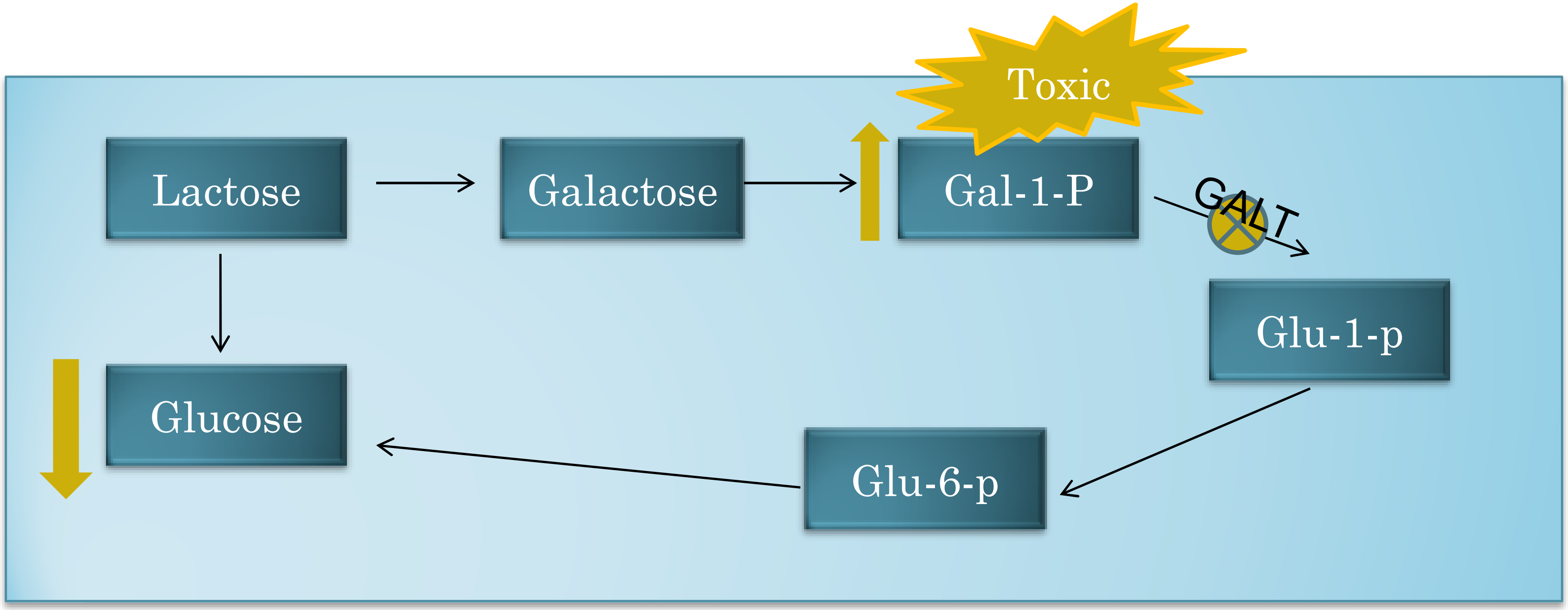
*Sanchez Russo RL & Wilcox WR.(2021) Amino Acid Metabolism. In Pyeritz RE et al. Emery and Rimoin's Principles and Practice of Medical Genetics and Genomics: Metabolic Disorders. Elsavier. 49-104

Treatment

- Nitrogen Scavengers
  - Sodium Phenylacetate/phenylbutyrate
  - Sodium Benzoate
- Low protein Diet
- Arginine supplementation
- Liver Transplant

Galactosemia

- Defect in the pathway of Galactose metabolism leading to accumulation of toxic Galactose-1-Phosphate



Galactosemia

- Presentation
  - Within the first 1-2 weeks of life
    - Poor feeding, vomiting, jaundice and lethargy.
    - Hepatomegaly, edema and ascites (Liver failure)
    - Cataract
    - Gram negative infection (E.coli)
    - Hypoglycemia
    - Renal tubular dysfunction

*Zenciroglu, A. et al. (2010). *European journal of pediatrics*, *169*(7), 903–906

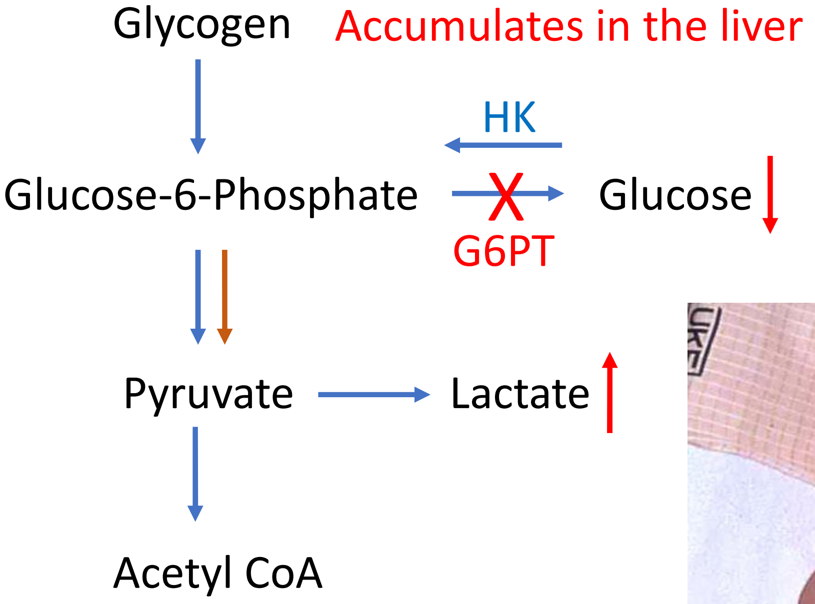
- Diagnosis
  - Urine reducing substance
  - Decreased GALT activity in blood
  - Increased Galactose-1-Phosphate
- Treatment
  - Galactose-free diet

Disorders of Energy Deficiency

- Glycogen Storage Disorders (GSD)
    - **Von Gierke Disease (GSD Type 1a)**
- Fatty Acid Oxidation Disorders (FAOD)
- Disorders of Ketogenesis and Ketolysis
- Oxidative Phosphorylation Disorders
    - **Mitochondrial Encephalopathy Lactic Acidosis Stroke Like Episodes (MELAS)**

Glycogen Storage Disease Type 1a

- Autosomal Recessive
  - Glucose 6 Phosphatase → *G6PT*
- Clinical symptoms:
  - Hepatomegaly
  - Hypoglycemia with lactic acidosis
- Treatment
  - Glucose
    - Frequent feedings
    - Complex carbohydrates/Corn starch
    - Overnight glucose drip
  - Liver transplantation



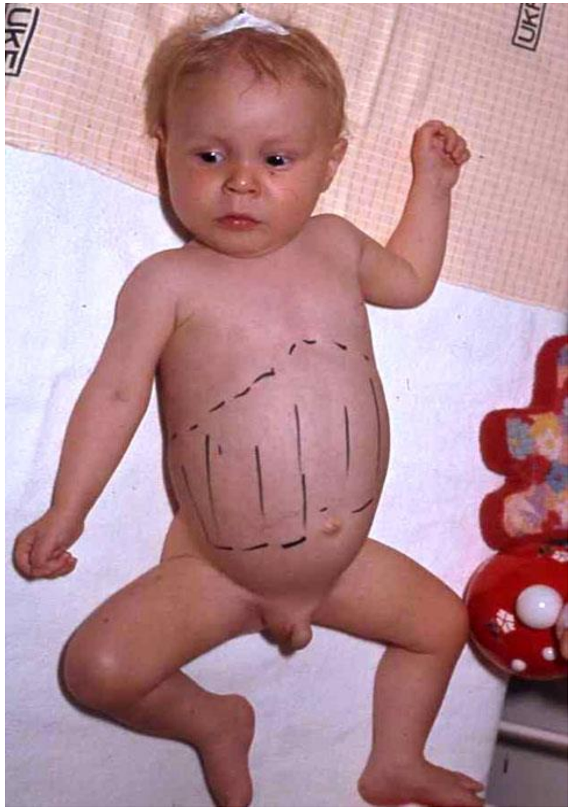
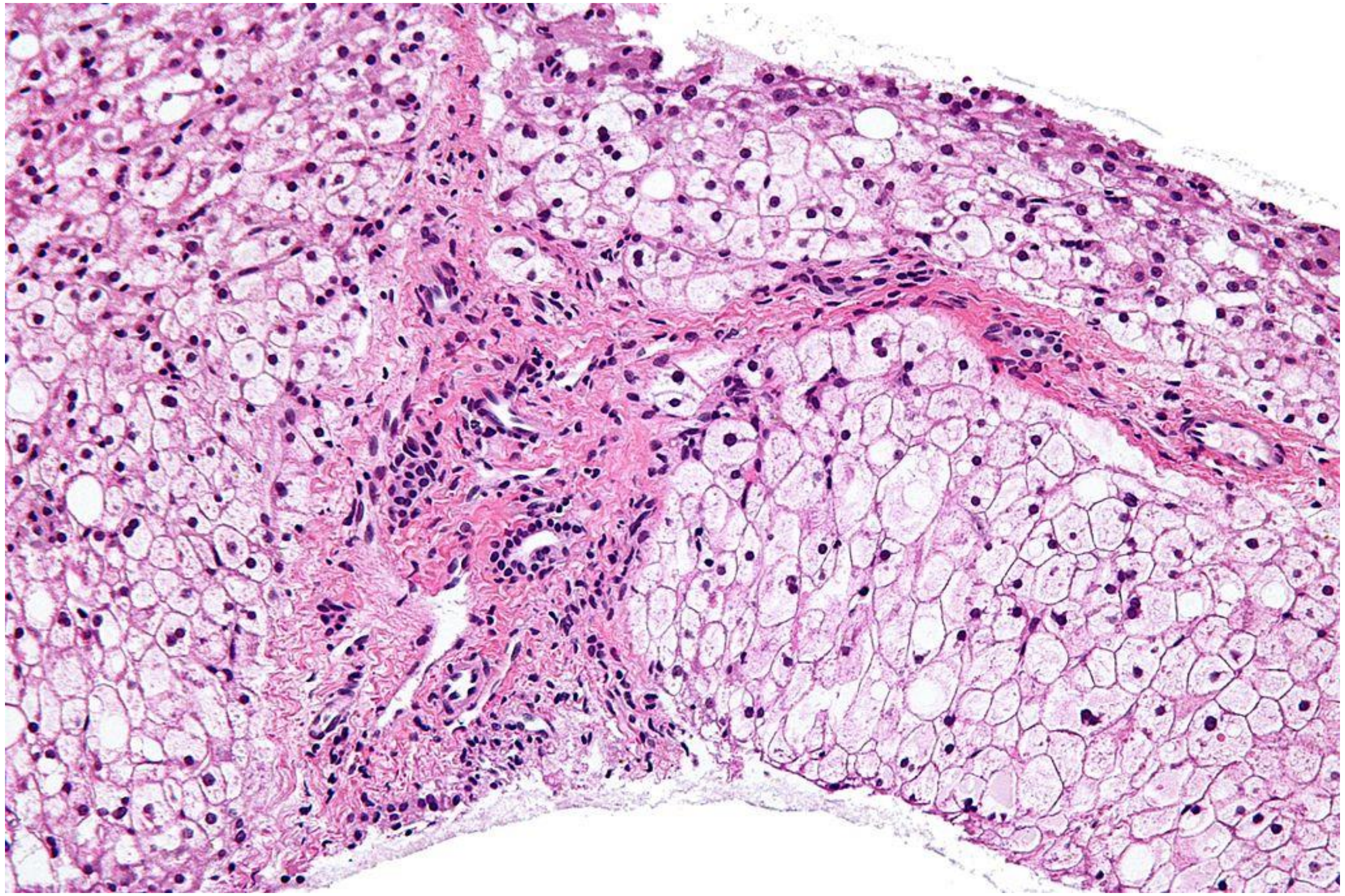


*Image from NAMA slides

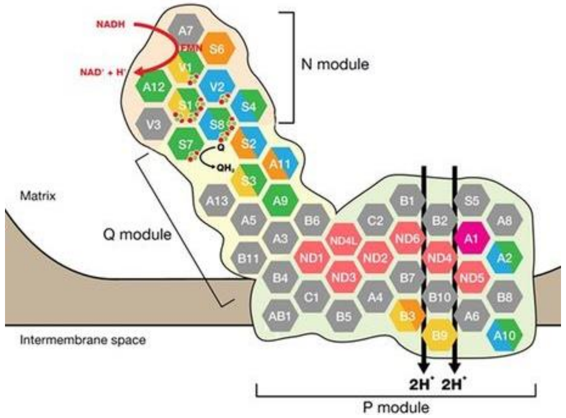
Biopsy of Liver from a patient with GSD Type 3



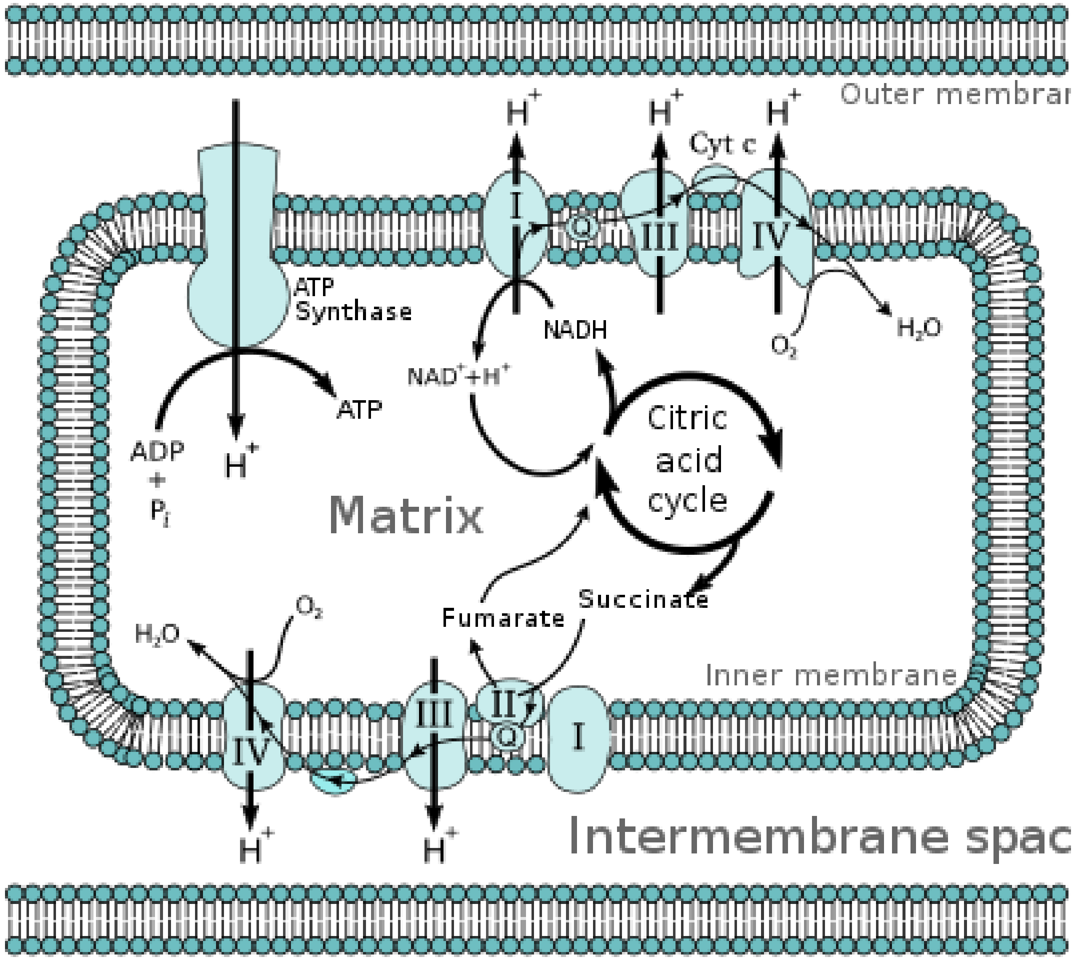
‘Glycogen Storage Disease’ (2021) Wikipedia. Available at:
https://en.wikipedia.org/wiki/Glycogen_storage_disease (Accessed: 5 January 2021).

Oxidative Phosphorylation – Mitochondrial

- Mitochondria
  - Powerhouse of the cell
  - Disorder of Energy Metabolism
- Inheritance
  - Maternally
  - Autosomal Dominant
  - Autosomal Recessive
  - X-linked



*Fassone, E., & Rahman, S. (2012). Journal of medical genetics, 49(9), 578–590.

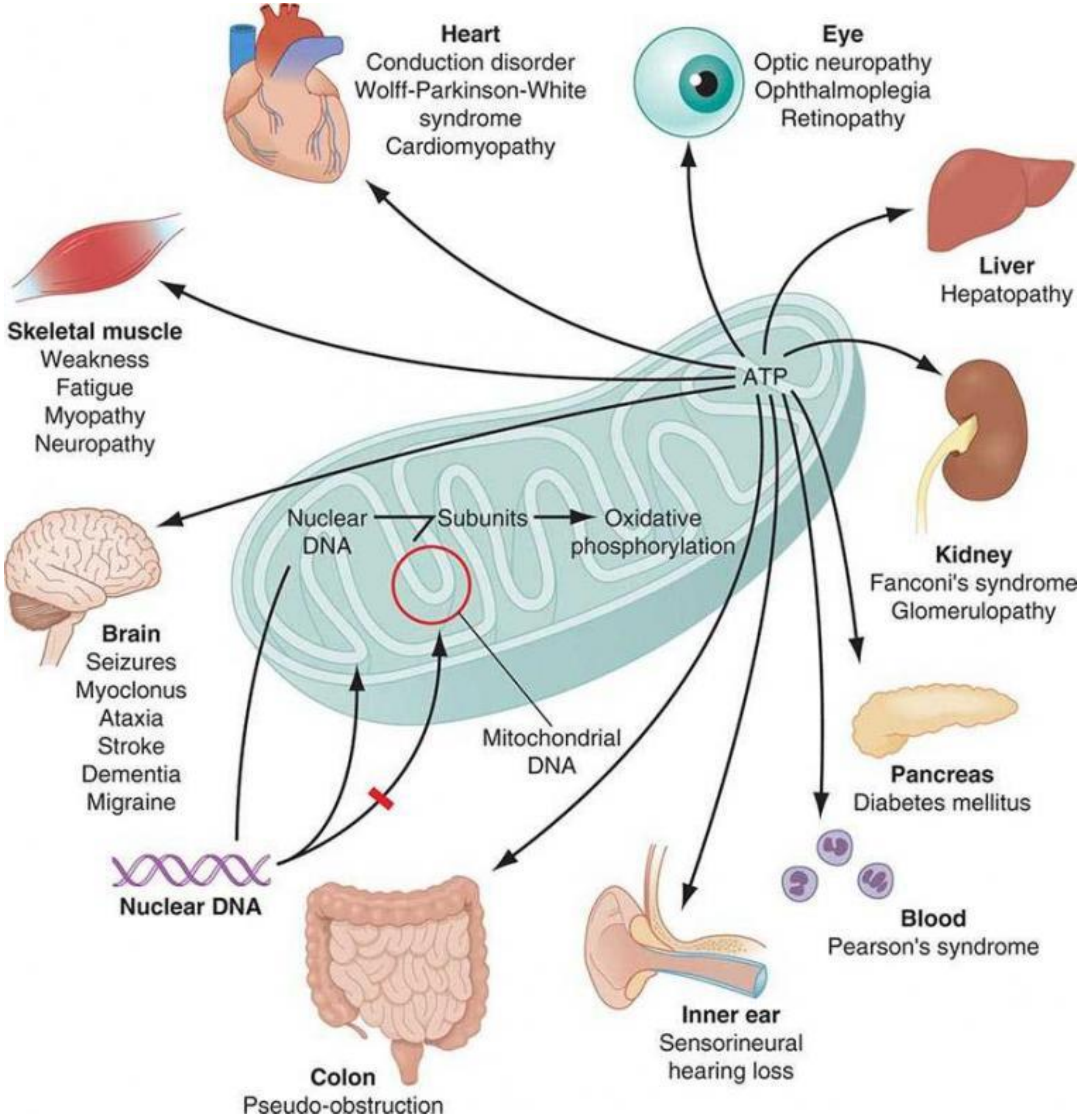


‘Electron transport chain’ (2021) Wikipedia. Available at: https://en.wikipedia.org/wiki/Electron_transport_chain (Accessed: 5 January 2021).

Clinical Manifestations

- Muscle weakness
- Cardiomyopathy
- Liver failure
- Seizure, encephalopathy, and stroke
- Deafness, and blindness

- Example:
    - MELAS syndrome (Mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes)



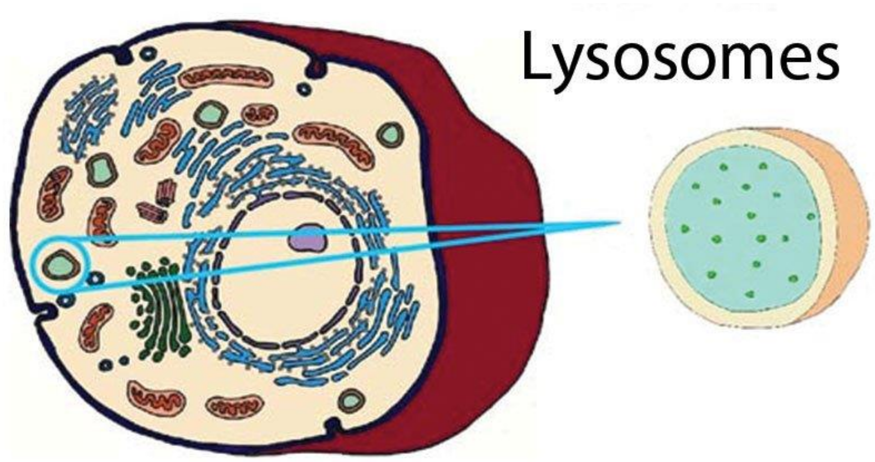
"Mitochondrial Disease" (2021) MitoCanada. Available at: https://mitocanada.org/mitochondrial-disease (Accessed: 5 January 2021).

Disorders of Complex Molecules

- Lysosomal Storage Disorders
  - Mucopolysaccharides (MPS)
    - **Hurler Disease (MPS1)**
  - Oligosaccharides
  - Sphingolipids
  - Sialic Acid Disorders
- Peroxisomal Disorders
  - **Zellweger Spectrum Disorder (ZSD)**
- Congenital Disorders of Glycosylation

Lysosomes

- Main function is to breakdown complex molecules
  - Contains Hydrolyzing enzymes
    - > 40 enzymes
    - Acidic PH



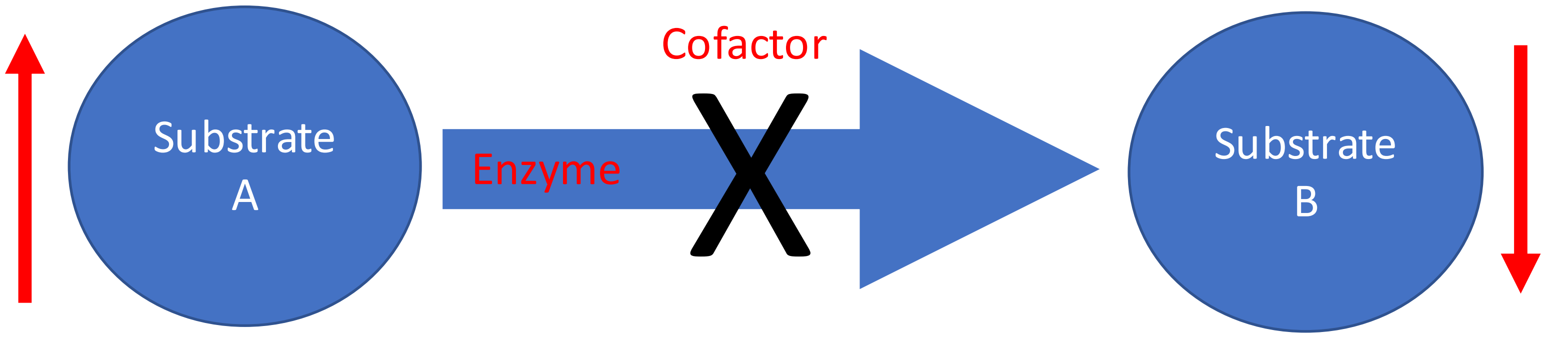
“Lysosomes- Definition, Structure, Functions and Diagram” (2021) Microbe Notes. Available at: https://microbenotes.com/lysosomes-structure-enzymes-types-functions (Accessed: 5 January 2021).

Hurler Syndrome

- Presentation
  - Intellectual Disability
  - Short stature
  - Coarse Facial features
  - Corneal Opacity (not present in Hunter syndrome)
  - Skeletal deformity
  - Hepatosplenomegaly
- Diagnosis
  - Increased Urine Mucopolysaccharides
  - Reduced enzyme activity
- Treatment
  - Enzyme Replacement Therapy
  - Hematopoietic Stem Cell Transplant

*Jones S & Wijburg F. (2016) Mucopolysaccharidoses, Oligosaccharidoses and Sialic Acid Disorders. In Saudubray JM et al. Inborn Metabolic Diseases. Springer. 577-590

Treatment of Inborn Errors of Metabolism



- Diet
  - Substrate reduction/restriction
    - Phenylalanine in PKU
  - Substrate supplementation
    - Glucose in GSD1a
    - Tyrosine in PKU
- Cofactor supplementation
  - Biopterin (BH4) in PKU
- Enzyme Replacement Therapy
  - Idursulfase in MPS1
- Organ Transplant
  - Liver transplant in GSD1
  - HSCT in MPS1
- Gene Therapy
  - Libmeldy for Metachromatic Leukodystrophy